

Personality Traits and Emotion Recognition to Predict Engineering Students' Soft Skills using applications of LLMs

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Background and aims

Soft skills matter in engineering education. This project investigates whether interview language and vocal delivery can support their estimation, using personality as an intermediate step.

Research question

Can LLM-derived text cues and audio characteristics predict interview ratings and support an exploratory Big Five-to-BESSI skill mapping?

Objectives

1. Extract structured language and audio features.
2. Compare predictions with simple baselines and published RecruitView results.
3. Evaluate a separate personality-to-skill mapping and identify its limits.

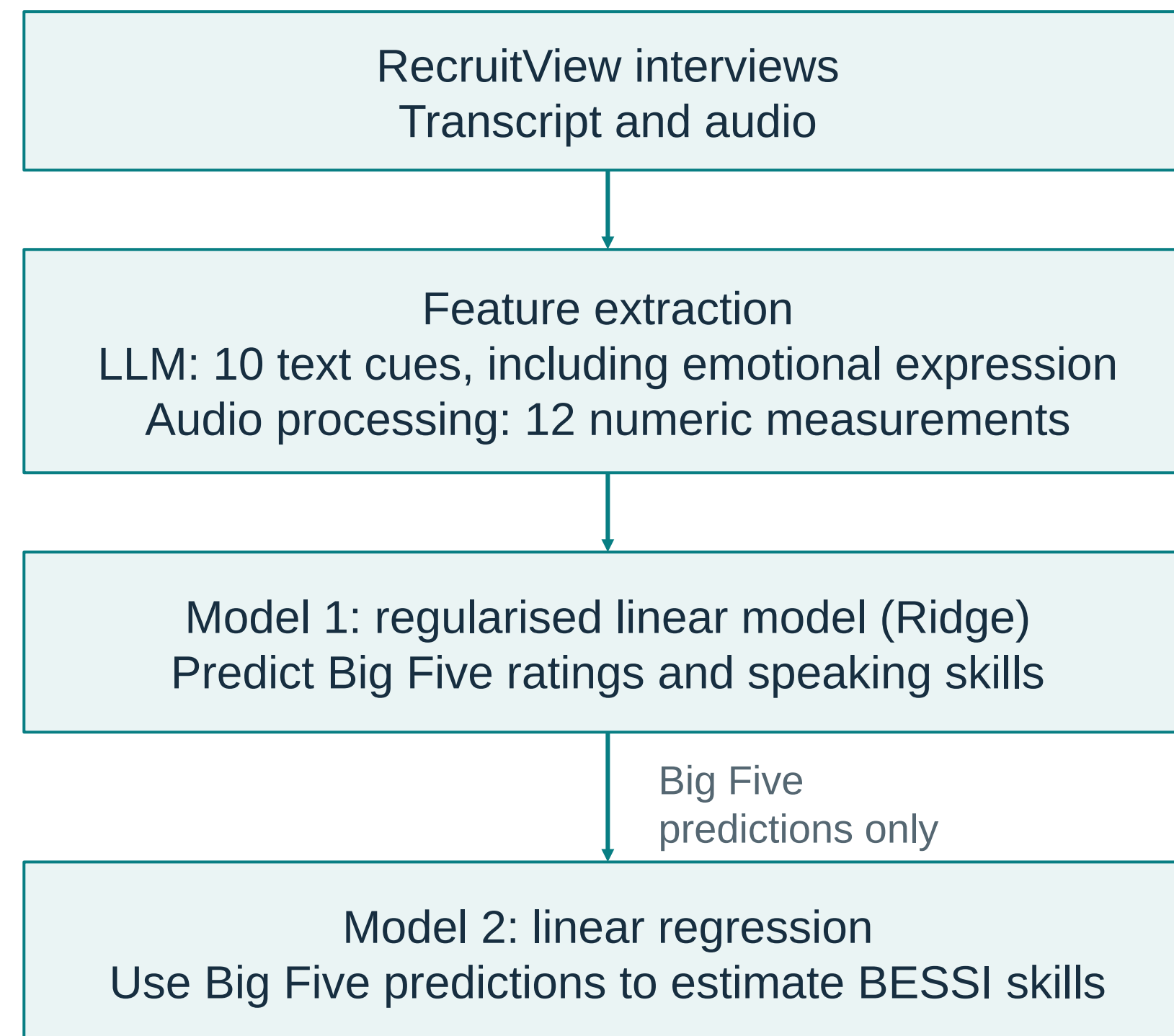
Research basis

RecruitView [1] provides interview personality and performance ratings. BESSI research [2, 3] studies related personality and skill measures. Relatedness supports testing a mapping, not assuming that personality and skills are equivalent.

Data

RecruitView: 1,998 responses, 331 people.
Paired Big Five/BESSI data: 313 usable rows.
No visual features in the final model.

Method and evaluation



Model 2 learns from separate paired questionnaire data. Statistical scale adjustment connects the two models. Speaking skills remains a separate Model 1 output.

Why this model?

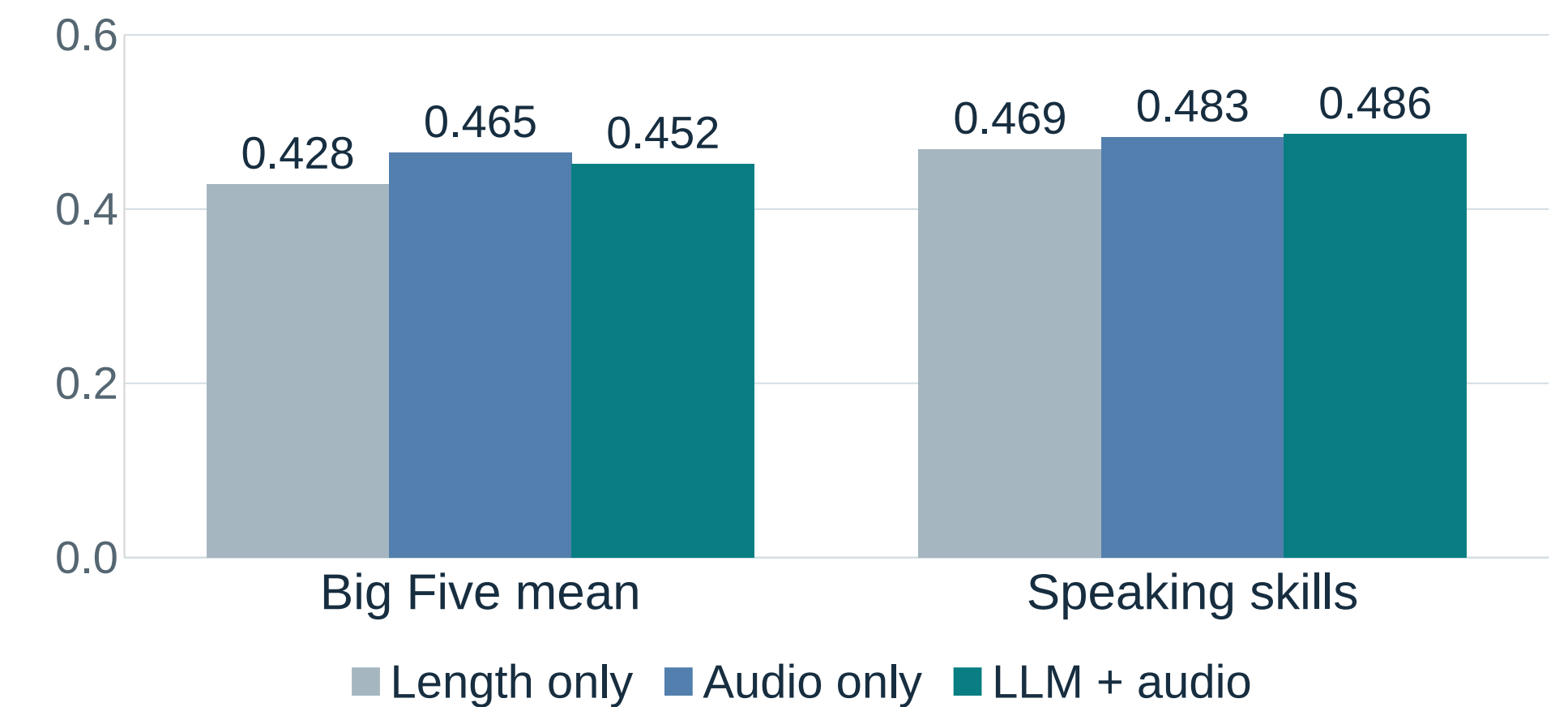
Ridge was selected using validation data. A gated neural follow-up did not improve Big Five and speaking skills consistently, so Ridge remains the primary model.

Evaluation checks

Keep each RecruitView participant in one split.
Choose the model on validation data. Check stability, response length and uncertainty.

Results and interpretation

Interview prediction: higher correlation is better



Spearman correlation, not percentage accuracy.
Test: 292 responses from 49 participants.

Published paper comparison [1]

CRMF: Big Five 0.523; speaking skills 0.595.

The paper also uses video and a different split. This is a contextual comparison, not a reproduction.

Separate BESSI mapping

Mean $R^2 = 0.504$ on 47 test rows. Participant independence is unverified. This is not the accuracy of the full interview-to-skills pipeline.

Main finding

The model finds moderate associations. However, the richer features did not clearly beat response length: the uncertainty range includes zero, so the small observed improvement may be sampling noise.

Project plan and current status

The chart shows the full 12-week project, not just poster preparation.
Current position: Week 6 (14–20 September 2026).
Completed: data, models, comparisons and final checks. Remaining: poster and dissertation work.
Poster deadline: 16 September 2026.

Limits: no direct BESSI labels in RecruitView; emotion cues lack independent validation; engineering-student generalisation is untested. Follow-up test analyses are exploratory. Research use only; do not use these estimates for hiring decisions.

[illegible]

References: [1] Gupta et al. (2025), RecruitView, arXiv:2512.00450. [2] Soto et al. (2022), BESSI, doi:10.1037/pspp0000401. [3] Sewell et al. (2022), social, emotional and behavioural skills, doi:10.1016/j.dib.2022.107792.